FRONTLINE MAPS

How Money and Rewards Work

A complete guide to the payments system: the Stripe building blocks, and the four reward areas built on them: Earn, Hire, Impact and Perks. Written for teammates, partners and advisors.

Earn

Sponsors fund cleanup pools. Cleaners and groups get paid for verified work.

Hire

Neighbors pay cleaners for jobs. Money is held, then released on confirmation.

Impact

One public page: money in, money out, and what it cleaned up. Plus the Frontline Plus subscription.

Perks

Partner business offers redeemed with points. No cash involved.

Version of 2026-11-01. Status: built and running on Stripe **sandbox** keys, visible to admins and the rewards beta list. Going live on real money is a separate, documented step.

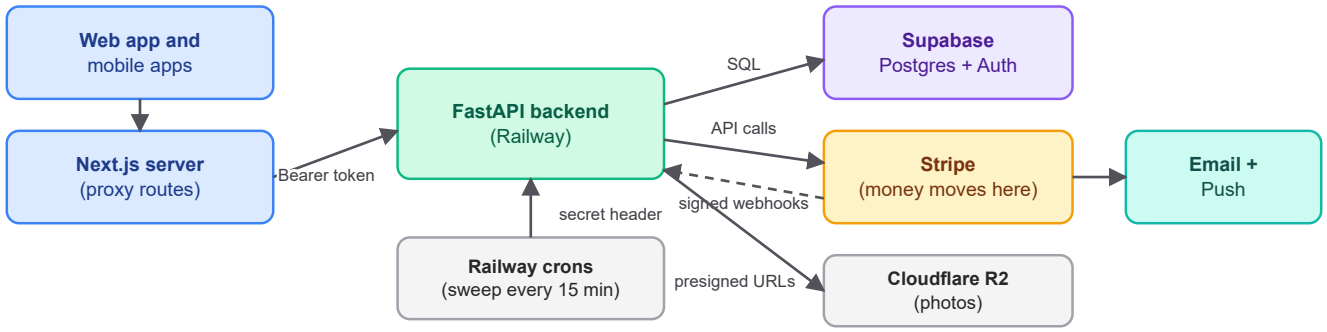
1. The whole thing in one page

- **Stripe moves all the money.** Our database only mirrors what Stripe tells us, through signed webhooks. We never store card numbers or bank details.
- **Four flows use Stripe:** Frontline Plus subscriptions, sponsorships (pools that pay Earn), Earn payouts, and Hire jobs. **Perks is the odd one out:** it spends points, not dollars.
- **People who get paid** (cleaners, groups, Earn workers) each have a **Stripe Connect Express** account. They add a bank or debit card on Stripe's own pages.
- **Earn:** sponsor pays into a pool, cleaner does the work with photos and GPS, an admin approves, and the transfer is automatic.
- **Hire:** the customer pays price plus a 5.5% fee on top when a cleaner takes the job. Money sits in the platform balance (the hold) until the customer confirms, or 3 days after the cleaner finishes.
- **Safety nets:** idempotency keys on every Stripe call, a 15 minute sweep job, strikes for cleaners who no-show, disputes that freeze money, and admin alerts for chargebacks and reversed payouts.
- **Access is gated** by three admin switches and a per-user beta flag, so real money can be tested in production by admins only.
- **Identity is proven by a signed token** (Supabase JWT) on every money route. The backend never trusts a user id sent by the browser.

2. The system at a glance

Every piece, and who talks to whom.

System map



Solid arrows are requests; the dashed arrow is Stripe calling us back. Photos upload straight to storage with a short-lived link. The browser never holds Stripe secrets; it only gets Stripe-hosted page links (Checkout, onboarding, billing portal) from the backend.

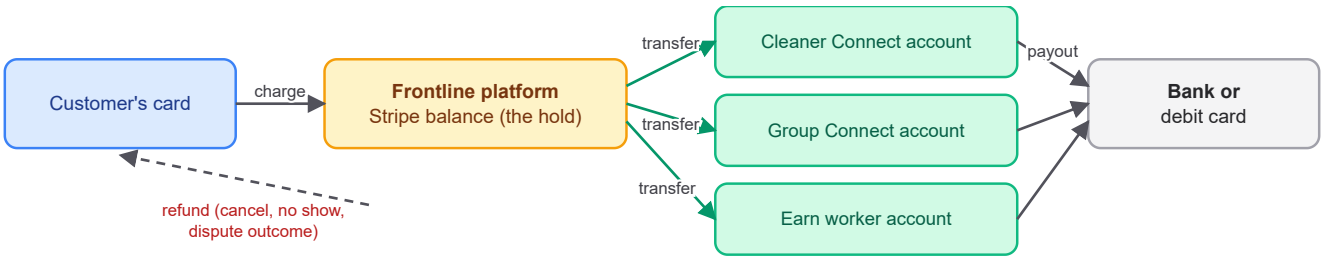
Piece	What it does	What it never does
Next.js app and proxies	Renders pages, forwards the signed-in user's token to FastAPI, hides the backend URL.	Does not decide who a user is by itself.
FastAPI (Railway)	All business rules: pricing, holds, strikes, approvals, transfers. Verifies the token on every money route.	Never trusts a user id sent from the client.
Supabase	Postgres data, user accounts and logins, row level security for direct browser reads.	Does not hold card or bank data.
Stripe	Charges cards, holds funds in the platform balance, pays out to Connect accounts, runs subscriptions, sends webhooks.	We do not copy its balances; we mirror its events.
Railway crons	The sweep (every 15 min) moves Hire jobs along. Other crons run reminders and point decay.	Cannot be called by the public; they send a shared secret.
Cloudflare R2	Stores before and after photos, site photos, proof images.	Photos are uploaded straight from the device with a short-lived signed link.
Email and push	Notifies on jobs, approvals, payouts, disputes and strikes.	Chat blocks phone numbers and links so payment stays on platform.

3. Stripe building blocks

Five Stripe features do all the work. Everything in Earn, Hire, Impact and Plus is a combination of these.

Stripe feature	Used for	Our table that mirrors it
Checkout (hosted page)	Plus subscriptions, sponsorships, saving a card for Hire.	`subscriptions`, `sponsorships`
PaymentIntent (off session charge)	Charging the customer's saved card the moment a cleaner takes a Hire job.	`marketplace_jobs` (payment intent id, `hold_status`)
Platform balance	The **hold**. Charged money sits here, not with the cleaner, until release or refund.	`hold_status`
Connect Express accounts	Where cleaners, groups and Earn workers get paid. Stripe collects identity and bank details.	`cleaner_profiles`, `payout_accounts`, `group_payout_accounts` (only the account id and `payouts_enabled`)
Transfers	Moving money from our balance to a Connect account: Hire release, Earn payouts.	`stripe_transfer_id` on jobs and allocations
Refunds	Returning the customer's money on cancel, no show or dispute resolution.	`hold_status = refunded`
Webhooks	Stripe tells us what happened. Signature verified; duplicate events ignored by event id.	`stripe_events`

Where the money physically goes



Customers and sponsors pay into the platform balance. Nothing reaches a payee until we create a transfer. Refunds go back to the original card.

Why the platform balance? Holding funds on the platform (a "separate charges and transfers" model) is what lets us refund a customer, freeze a disputed job, or wait for admin approval before anything leaves. Stripe fees come out of the charge, which is why the Hire fee is set above break even (section 7).

4. Who can see what

Three switches plus one per-user flag decide visibility. This is how real money is tested in production without the public seeing it.

Visibility controls

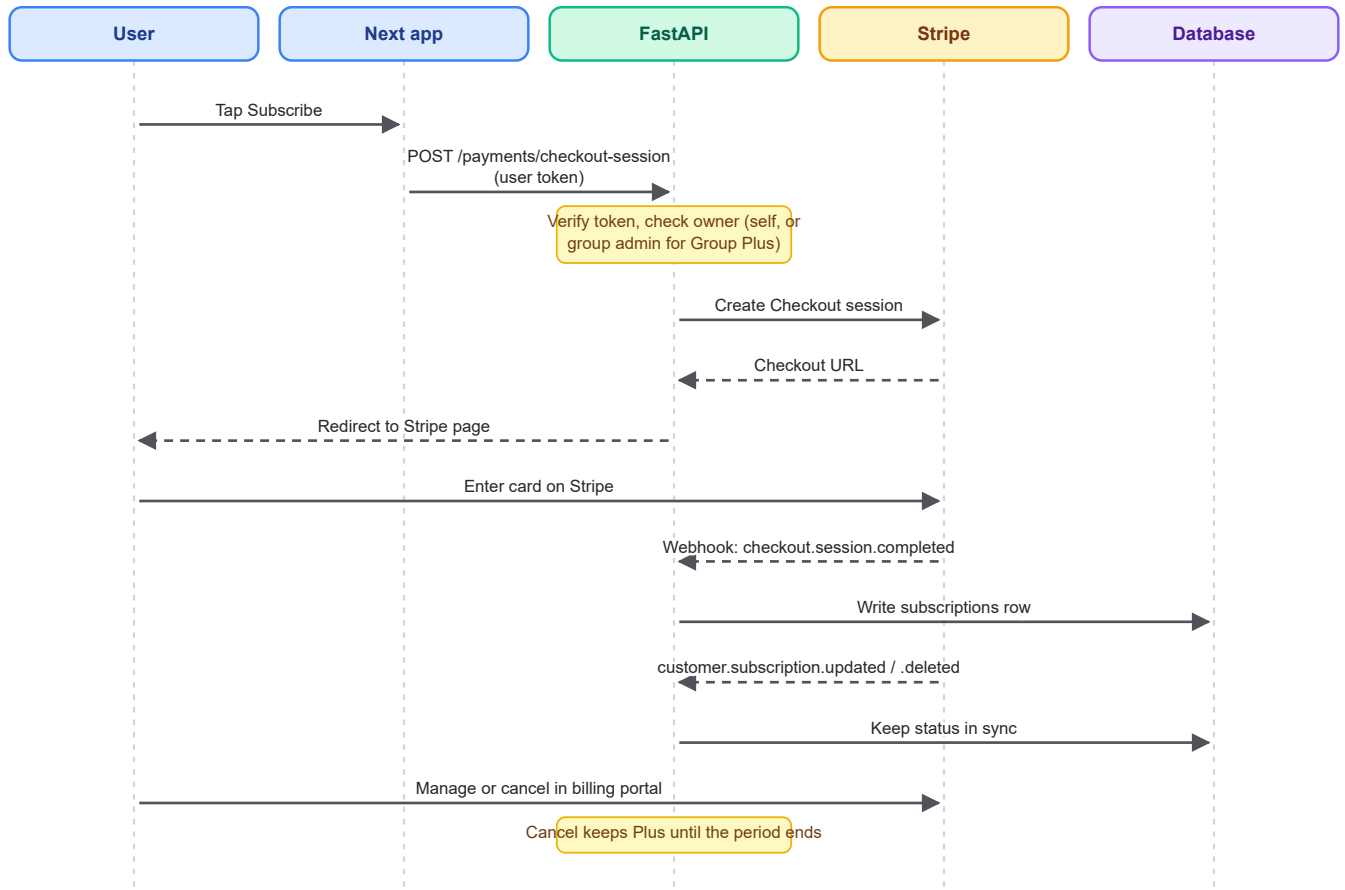


Page	Gate used	Notes
Earn, Impact, Hire	payments switches or `rewards_beta`	Enforced in the Next layer (page and proxy) and identity checked again in FastAPI.
Perks	payments switches	Points only, but gated with the rest of the rewards area.
Frontline Plus billing	payments switches	Subscription widget on the Impact page.

5. Frontline Plus (subscription)

$5 a month, for a person or for a group. Feeds Impact's supporter numbers.

Subscribe and stay in sync

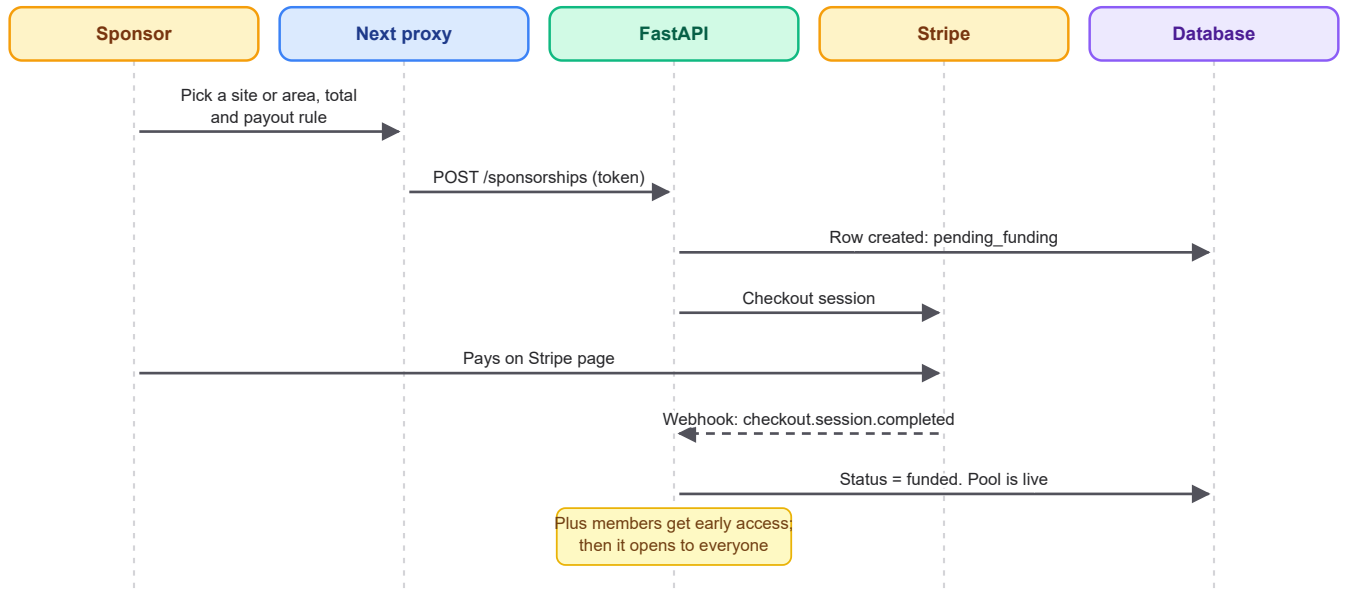


- The app only reads subscription **status** (active or trialing means Plus). It never trusts the redirect back from Stripe.
- **Perks of Plus:** early access to newly funded sites, a higher Earn cap (a multiplier set in Settings), the monthly recap, a supporter badge and insights.
- Entitlement lookups return only a yes/no premium flag to other users; flair stays public.

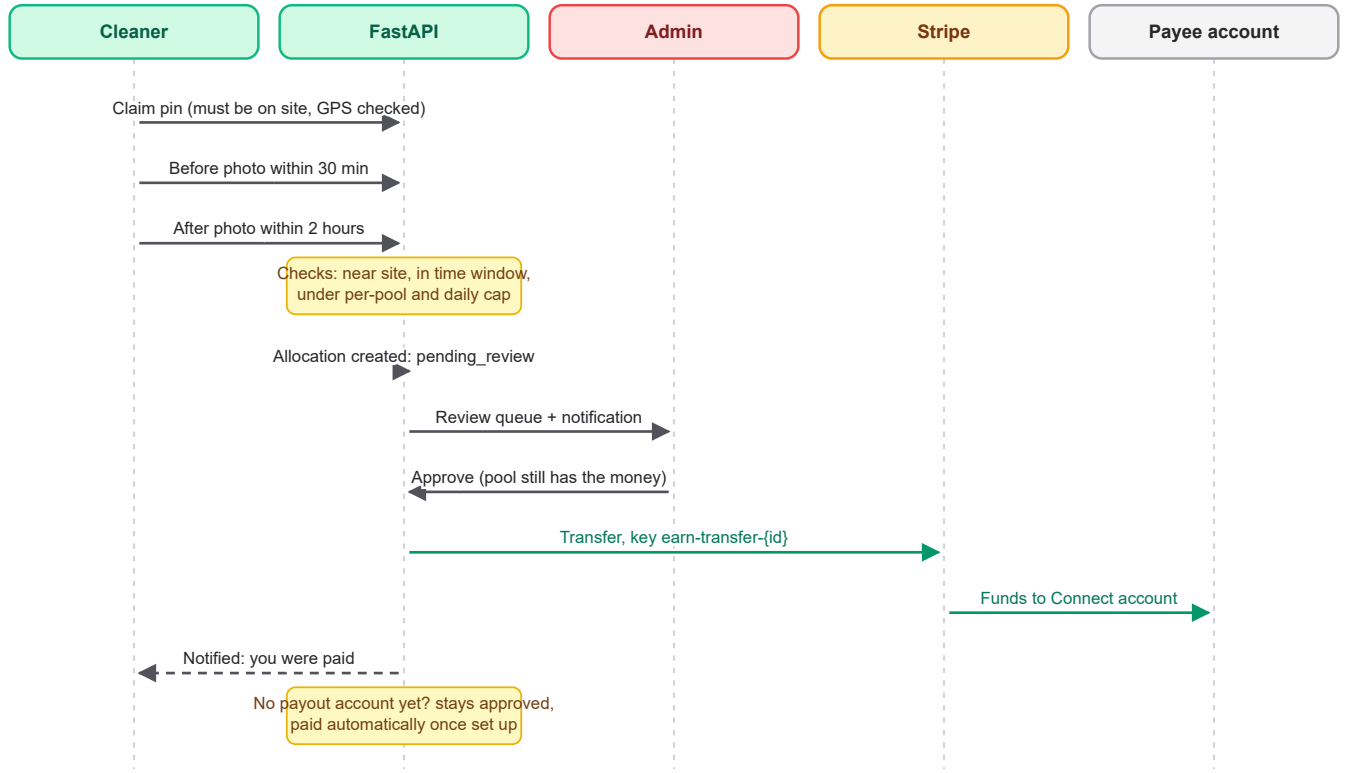
6. Earn: sponsor funded cleanup rewards

A sponsor funds a pool. Verified work draws payments from it. An admin approves each payout.

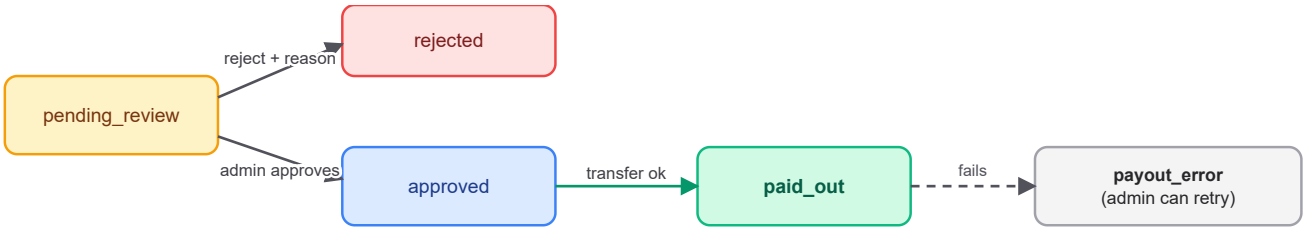
Step 1: funding a pool



Step 2: earning, approval and payout



Allocation states



Group allocations (no single person to pay) use a manual "Mark paid out" flag. Individuals are paid by Stripe transfer.

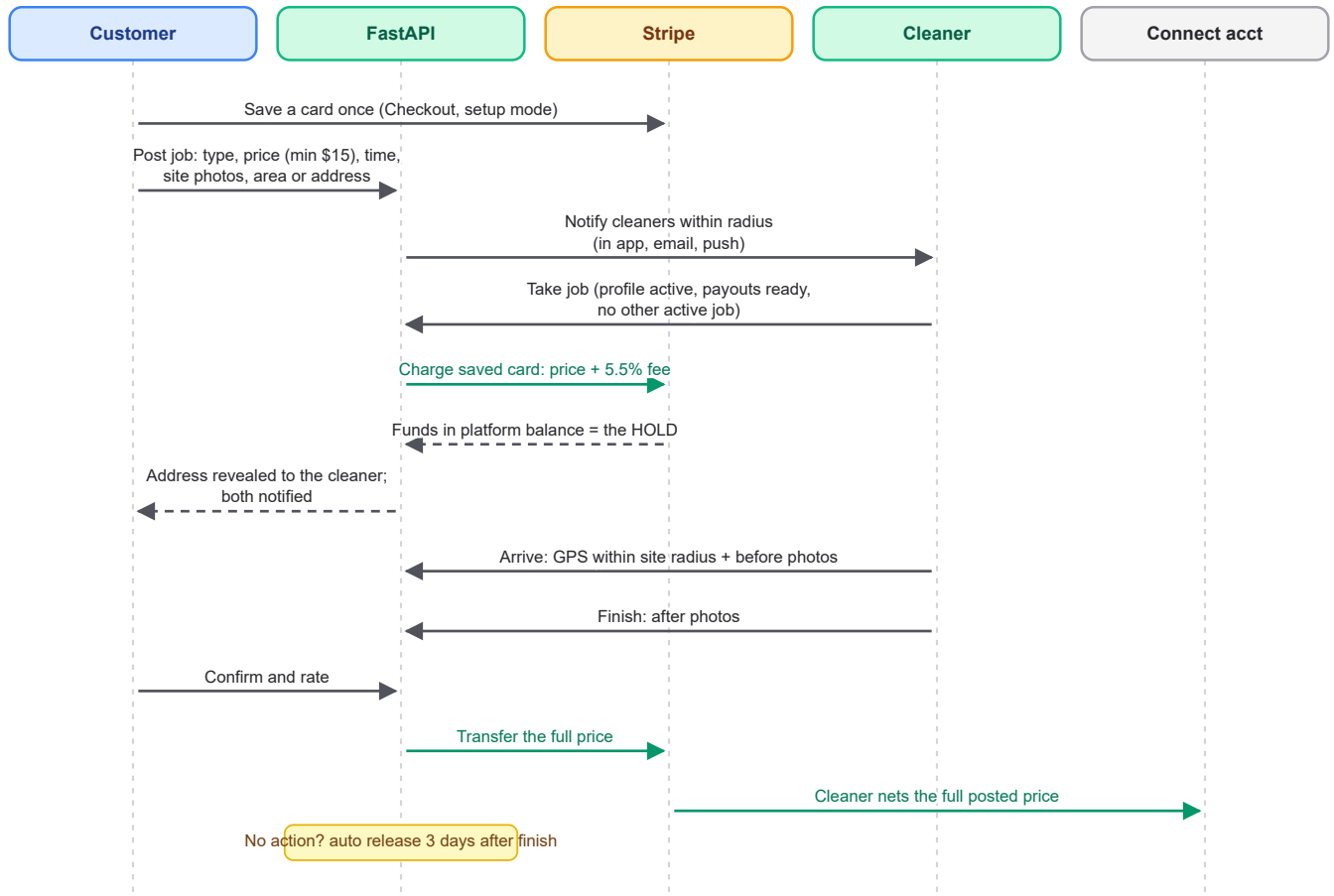
Payout rule	How it pays
One payout	First approved cleaner takes the whole pool.
Fixed per cleanup	Same amount for each approved cleanup. The last payout gets the remainder.
Per bag	Amount times bags logged and verified.
Per attendee (areas only)	A group schedules an event in the area, marks it sponsored; verified attendance pays the group.

Earn guardrail	Detail
Presence	Cleaner must be near the site; before photo 30 min after claim, after photo 2 hours.
Time out	Running out of time reopens the site for others.
Caps	Per pool and per day per person (Settings > Payments > Earn). Plus raises the caps.
Manual approval	Every allocation needs an admin today. Auto-approve for trusted cleaners is scoped, not built.
Idempotency	One transfer per allocation, ever (key earn-transfer-{id}).
Missing payout setup	Allocation waits as approved; a group without payout setup cannot host a sponsored event.

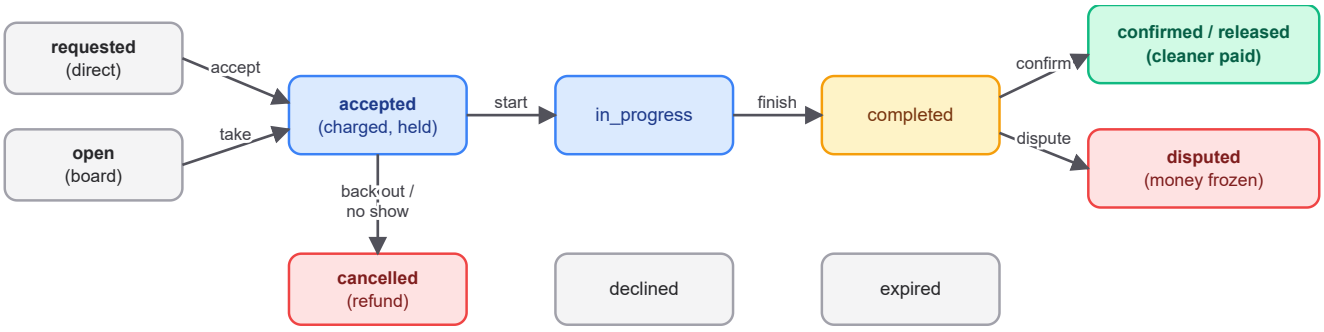
7. Hire: paid cleaning jobs

A customer posts a job at a price. The first cleaner to take it gets it. Money is held until the work is confirmed.

Happy path, end to end



Job lifecycle



Back out or no show on a board job returns it to open (customer refunded, charged again to the next cleaner). Direct jobs are cancelled and refunded. After the job starts only the cleaner can back out (a strike). The customer opens a dispute instead.

Where each dollar goes

Worked example: $50 job at the default 5.5% fee

Customer is charged $52.75 (price $50.00 + Frontline fee $2.75)

Cleaner receives $50.00 (full posted price) | **Stripe fee** $1.83 | Frontline keeps $0.92

Stripe charges 2.9% + $0.30 on the card total ($1.53 + $0.30). The fee is paid ON TOP by the customer, so the cleaner is never short.
At the $15 minimum job, about 5.05% covers Stripe, so 5.5% breaks even and earns a little on larger jobs. Settings has a fee check tool.

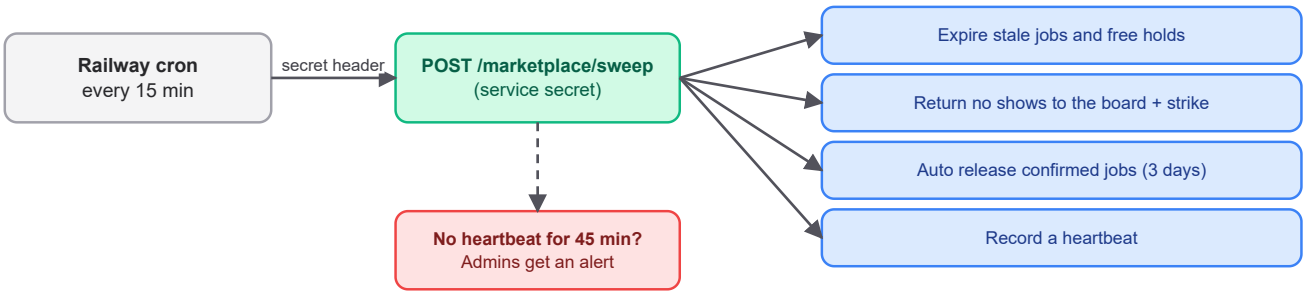
Protection	How it works
One active job	A cleaner can hold one job at a time.
Start-by time	Missing it returns the job to the board and gives the cleaner a strike.
Strikes	Back out after the start time or once work has begun, or a no show = 1 strike. 3 in 90 days suspends the profile until an admin reinstates. The cleaner is notified and the app shows a red suspended banner.
Presence proof	Arrive requires GPS within the site radius (a setting) and before and after photos.
Address privacy	Board listings coarsen the pin and hide address, area and radius. The full address shows only after a cleaner takes the job.
Chat safety	Messages with phone numbers, emails or links are blocked to keep payment on platform. Abusive messages go to a Flagged messages queue.
Disputes	Opening one before release freezes the money. An admin reads chat and photos, then refunds the customer or releases to the cleaner.
Idempotency	Every Stripe charge, refund and transfer has a key, so retries cannot double charge or double pay.
Legal gates	Customers and cleaners accept an agreement first; cleaners are independent contractors.

8. The engine room: webhooks and the sweep

Webhook event	What we do
`checkout.session.completed`	Activates Plus, marks a sponsorship funded, or saves a Hire card.
`customer.subscription.updated / deleted`	Keeps Plus status in sync.
`invoice.payment_failed`	Logged for follow up.
`charge.refunded`	Marks a sponsorship refunded.
`account.updated`	Keeps a payee's `payouts_enabled` flag in step with Stripe.
`charge.dispute.created`, `transfer.reversed`	Money may have left. **Every admin gets an alert.** Review in Stripe before any related payout moves.

One endpoint handles all of them: `POST /api/payments/webhooks/stripe`. It is the only money route that uses Stripe signature verification instead of a user token, and duplicate deliveries are ignored by event id.

The sweep

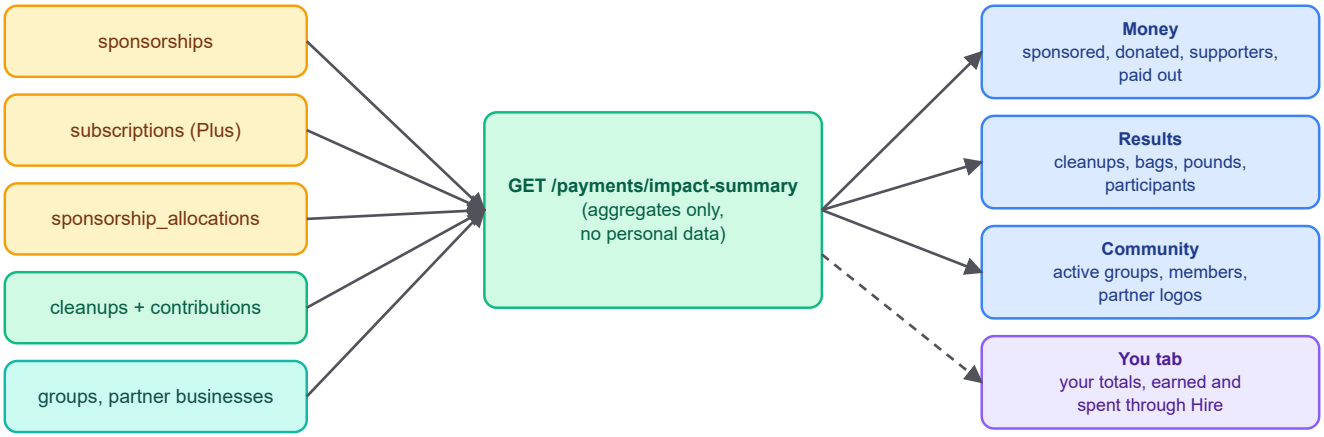


If the sweep stops, jobs stop moving. The alert fires at most once per 6 hours.

9. Impact: the transparency page

A read-only summary of money in, money out and real world results. Also hosts the Plus subscription widget.

Where Impact numbers come from

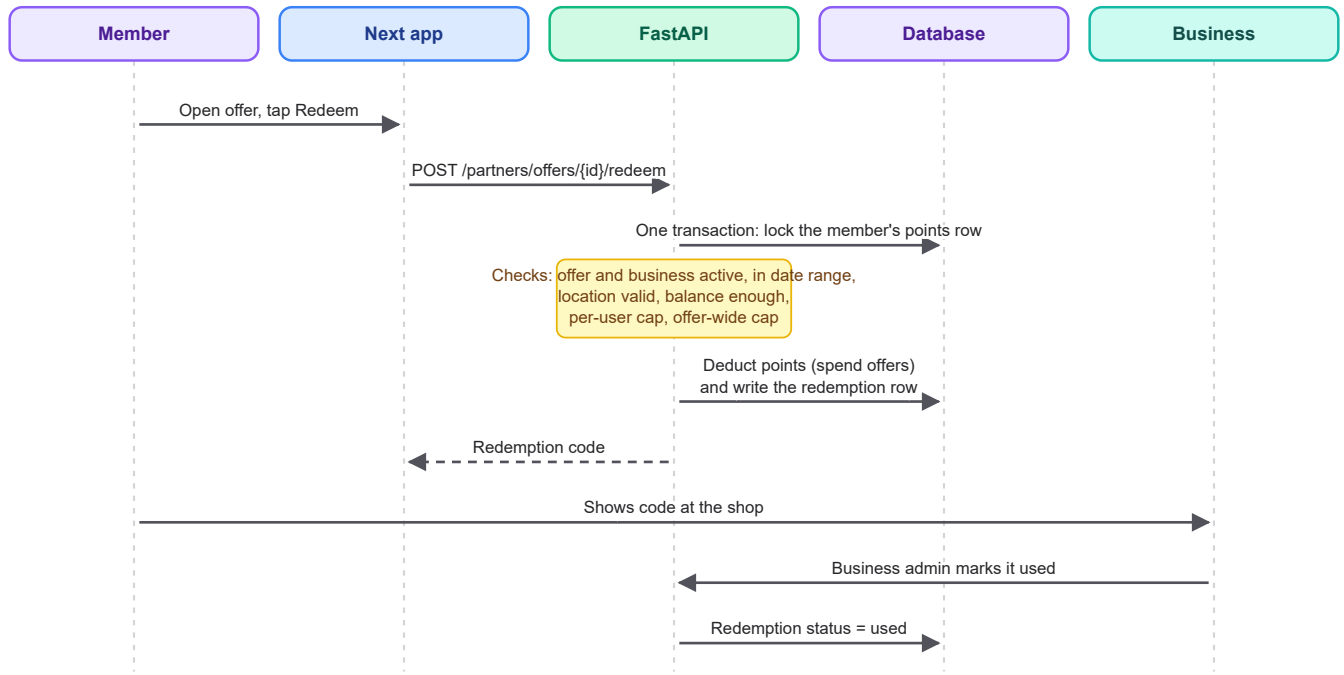


- **Counting rules:** a sponsorship counts once it is funded, released or exhausted. "General" gifts to Frontline are shown as donations and are separate from site or area sponsorships. A cleanup counts only once it has really happened (completed, verified, or a past event with logged totals).
- **Paid out vs owed:** the ledger separates approved-but-unpaid from paid-out allocations.
- **Monthly chart:** 12 months of gifts and new supporters.
- **You tab:** personal totals from your own contributions, plus Hire money earned (paid and pending) and spent.
- **Privacy:** the public summary is aggregate only. The Plus entitlement call exposes only a yes/no flag.

10. Perks: partner offers bought with points

No dollars move. Points earned from cleanups are spent at local businesses.

Redeeming an offer

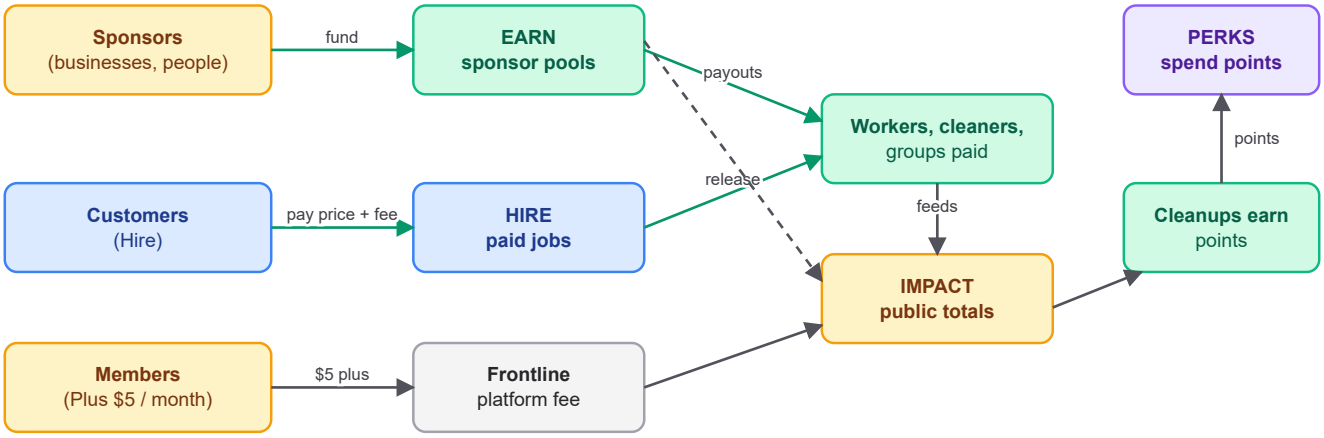


Concept	Detail
Offer modes	**Spend**: points are deducted on redeem. **Threshold**: unlocked at a points level, nothing deducted.
Limits	Per-user redemption cap, offer-wide total cap, start and end dates, optional single location.
Event offers	Some offers attach to a cleanup event so attendees can claim them; businesses get notified of the attachment.
Business side	Business admins manage offers, locations and other admins, and mark redemptions used.
Atomic	Balance check and deduction happen in one database transaction with the row locked, so a double tap cannot overspend.

Security status: Perks is the next router in the backend security rollout. Today its redeem route still takes the user id from the request; the plan moves it behind the same signed token check used by Earn, Hire and Impact (see section 12). Perks is the only reward area that has not been migrated yet.

11. How the four areas connect

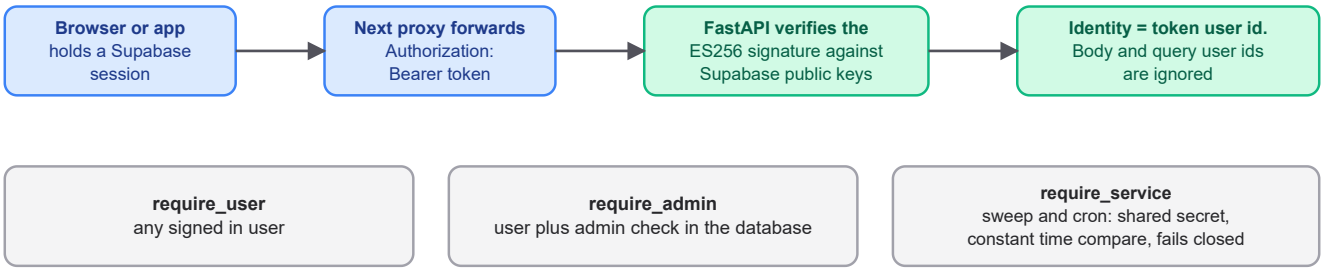
Money and points across the product



Green arrows are cash. Earn and Hire both pay people; Plus and fees fund Frontline; Impact reports all of it; Perks is a separate points economy fed by the same cleanups.

12. Security model for money routes

Every money route proves who is calling



- Expired, wrong audience, wrong issuer and HS256 tokens are rejected. Owner checks (self, group admin, site admin) live in the service layer.
- Money and write routes are rate limited: 10 per minute for money actions, 60 per minute for other writes, per user.
- A test walks every route under Hire, Earn and billing and fails if one lacks an auth dependency. Only the Stripe webhook is exempt (it uses its signature).
- CORS methods and headers are narrowed. The Stripe webhook never uses the shared secret.
- **Next:** the same pattern is being rolled across the rest of the backend, in priority order, tracked in `backend-security-plan-2026-10-30.md`.

13. Admin day to day

Task	Where and how
Approve Earn work	Admin shows sponsored cleanups and events waiting. Approving checks the pool, then pays automatically.
Resolve a Hire dispute	Read chat and photos, then refund the customer or release to the cleaner.
Reinstate a suspended cleaner	From the cleaner list after a conversation. The cleaner is notified.
"I was not paid"	Check the person's `payouts_enabled` first. Unfinished Stripe onboarding means nothing can transfer. Then check Transfers in Stripe.
Refund	Issue it in the Stripe dashboard. The webhook updates our rows.
Alerts	Chargebacks, reversed payouts and a stalled sweep arrive as admin notifications.
Settings	Fee percent, minimum job, site radius, Earn caps, Plus multiplier, and the visibility switches.

14. Where things stand, and what is left

Area	Status
Plus, Earn, Hire, Impact	Built. Running on Stripe sandbox keys in production for admins and beta testers.
Backend auth for these four	Done for Earn, Hire and Impact/billing. Perks and the rest of the backend are queued.
Going live	A runbook covers Connect, webhooks, keys, sandbox cleanup and a live test. Not yet executed.
Mobile return from Stripe	Scoped, not built: after Stripe pages, phones open the browser instead of the app.
Plus in native apps	Needs an in-app purchase decision (app store rules).
Earn approval automation	Scoped, launching with manual approval.
Debit-card-only onboarding	Untested.

Glossary

Term	Meaning
Hold	Money charged to the customer that sits in the platform Stripe balance until released or refunded.
Connect Express	A Stripe account type for payees; Stripe handles identity and bank details.
Transfer	Moving money from the platform balance to a payee's Connect account.
Pool	A sponsor's funded budget for a site or area, drawn down by approved allocations.
Allocation	One payout entitlement from a pool, awaiting approval, then payment.

Term	Meaning
Strike	A penalty for a cleaner no show or late back out. Three in 90 days suspends.
Idempotency key	A unique id on a Stripe call so retrying it can never repeat the money movement.
Sweep	The 15 minute job that expires, returns, strikes and releases Hire jobs.
Webhook	A signed message from Stripe telling us something happened.

Source docs: payments-admin-guide-2026-11-01, stripe-go-live-runbook-2026-11-01, paid-cleaner-marketplace-scoping-2026-10-22, payments-v2-scoping-2026-10-15, backend-security-plan-2026-10-30.